Tan Zhe Kai

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Education

Nanyang Technological University, Singapore Aug 2023 – Dec 2026 (Expected)
Bachelor of Computing (Hons) in Computer Science

- Specialized in Artificial Intelligence.

Ngee Ann Polytechnic, Singapore Apr 2018 – May 2021
Diploma in Financial Informatics

- Specialized in Financial Analytics; Director's List for 3 semesters.

Experience

Seagate Technology | *Data Analytics & Automation Intern* Jun 2026 – Dec 2026

- Automated daily extraction of equipment telemetry from live production servers via scheduled Python jobs, cleaning and restructuring ∼1M records/day into DuckDB and Parquet stores and cutting data-prep time ∼85%.
- Built a Streamlit analytics dashboard on that pipeline, deployed on an on-premise host over the internal network, giving 20+ equipment engineers same-day performance visibility in place of manual Excel pulls.
- Developed modular tools for yield analysis, component changeout monitoring, low-yield unit detection and multi-file Excel consolidation, cutting anomaly time-to-detection from days to under an hour to protect production yield.
- Prototyping an agentic RAG assistant over the same data layer: an n8n-orchestrated reason–act–observe loop with schema-aware SQL tool-calling on DuckDB, vector retrieval over engineering SOPs, and scoped tool permissions.
- Targeting self-serve natural-language querying for 20+ engineers to remove ∼70% of recurring ad-hoc data-pull requests, with grounded responses evaluated against a golden set of real engineer questions.

Rolls-Royce Solutions Asia | *Order Management Intern* Jul 2025 – Dec 2025

- Automated SAP order-data processing with Excel VBA, cutting manual handling time ∼60% and eliminating recurring transcription errors from monthly reporting.
- Supported SAP S/4HANA UAT by documenting 30+ test cases, validating end-to-end order workflows, and resolving system defects via SAP Cloud ALM.
- Reconciled customer order data across SAP and multiple Excel sources, improving monthly delivery-planning accuracy and reducing reporting discrepancies and downstream rework.

The Coca-Cola Company | *IT Intern* Sep 2020 – May 2021

- Built Microsoft PowerApps applications that replaced paper-based processes, including a COVID-19 attendance and temperature-screening tracker delivered with the HR team.
- Managed technical device and asset inventory at 100% tracking accuracy and executed periodic data migrations with zero downtime.

Projects

SlideScribe – AI-Enhanced Speech-to-Text Presentation Assistant | *Python, Streamlit, WebRTC, Whisper, LLM APIs*

- Building a real-time speech-to-text pipeline that streams microphone audio over WebRTC, cutting chunks at detected pauses and gating out silence to remove hallucinated filler text and ∼30% of redundant API calls.
- Engineered a provider-agnostic LLM connector (Groq, OpenAI, Gemini) with ranked model fallback and automatic retry on transient API failures, keeping the app running through three upstream model retirements.
- Designed a positional edit-markup scheme for LLM grammar correction that preserves each error's location, driving a before/after review UI with per-correction accept/reject before python-pptx writes the notes into the .pptx.
- Benchmarking streaming vs. single-pass transcription on identical audio to quantify the accuracy–latency trade-off, alongside grammar-correction precision, for the project's evaluation study.

Autonomous Mobile Robot – Integrated Image Recognition | *Raspberry Pi, STM32, Python, Android Studio*

- Configured low-latency Bluetooth and Wi-Fi communication between Raspberry Pi and STM32, enabling reliable real-time robot control with sub-second response times.
- Integrated computer-vision icon recognition (90% accuracy) and shortest-path navigation with the Android and algorithm teams to complete the autonomous run.

Skills

Languages: Python, Java, C++, C#, SQL, JavaScript, TypeScript, MATLAB, R, Go, VBA, Bash, HTML/CSS
Data & ML: pandas, NumPy, scikit-learn, PyTorch, TensorFlow, DuckDB, Parquet, Streamlit, Power BI, Tableau, Plotly
Agentic AI: RAG & vector retrieval, tool calling, MCP, ReAct loops, context engineering, guardrails, agent evals, n8n
Microsoft & Cloud: Azure, Azure DevOps, Power Apps, Power Automate, SharePoint, Excel/VBA
Technologies: CI/CD, Git, Docker, Agile, React.js, Node.js, Firebase, Android SDK, SAP S/4HANA, Jupyter
Spoken Languages: English, Chinese, Malay (Basic)